
Scientific Watch Report agents ReAct+Reflexion

“fake news detection”

Generated on	14 July 2026 at 04:13
Topic	fake news detection
Articles fetched	50
Articles selected	15
Summaries produced	15
Reflexion iterations	1 / 3 max
Critic evaluations	1
Final synthesis quality	GOOD
Min. quality threshold	GOOD
Total tokens used	37,043

1. Trends, Gaps & Future Perspectives

Research Trends

Trend	Maturity	Description	Evidence
Multimodal and Cross-Modal Inconsistency Detection	ESTABLISHED	Research is increasingly moving beyond text-only analysis to integrate visual, social context, and user comment features. A key focus is on detecting cross-modal inconsistencies—deceptive connections between text and images—to mimic human cognitive reasoning for identifying fake news.	openalex_W4409283534, openalex_W4382239959, openalex_W4403431661
Transformer and Hybrid Deep Learning Dominance	ESTABLISHED	Transformer-based models (e.g., BERT, RoBERTa, CT-BERT) and hybrid architectures (e.g., CNN-LSTM, SA-BiLSTM) have become the standard, consistently achieving state-of-the-art results. These models excel at capturing complex semantic and syntactic nuances, often reaching near-perfect accuracy on benchmark datasets.	openalex_W4409283534, openalex_W4393170771, openalex_W4377115522
Explainable and Cognitive-Inspired AI	EMERGING	There is a growing emphasis on model interpretability and trustworthiness through Explainable AI (XAI) techniques like LIME and LDA, and cognitive-inspired frameworks. These approaches aim to make model decisions transparent and align detection processes with human reasoning patterns.	openalex_W4393170771, openalex_W4378697009, openalex_W4409283534
Large Language Models as Advisors	EMERGING	Instead of using Large Language Models (LLMs) as standalone detectors, a new trend uses them as advisors to smaller language models (SLMs). LLMs provide multi-perspective rationales that guide SLMs, combining strong reasoning capabilities with efficient, specialized detection.	openalex_W4393160654
Graph-Based and Social Context Modeling	EMERGING	Graph Neural Networks (GNNs) are being leveraged to model the propagation of fake news through social networks. Novel strategies like Bayesian convolution and long-tail user matching are improving robustness against data perturbation and the influence of spam bots.	openalex_W4390547902, openalex_W4384080794
User and Emotion Feature Integration	ESTABLISHED	Detection models are increasingly incorporating user-centric features (e.g., user profiles, social capital) and affective signals (e.g., sentiment, emotion from comments). This approach recognizes that user behavior and emotional reactions are strong indicators of news veracity.	openalex_W4319320113, openalex_W4384080794, openalex_W4378697009

Emerging Trends — Supporting Articles

[EMERGING] Explainable and Cognitive-Inspired AI

Advancing Fake News Detection: Hybrid Deep Learning With FastText and Explainable AI (2024)

The widespread propagation of misinformation on social media platforms poses a significant challenge, requiring robust detection solutions that can effectively identify media-rich fake news by leveraging multimedia elements and enhancing detection accuracy. A hybrid deep learning approach integrating FastText word embeddings with Machine Learning and Deep Learning models, including Convolutional Neural Networks and Long Short-Term Memory networks, enhanced with regularization and hyperparameter optimization. Transformer-based models (BERT, XLNet, RoBERTa) were also employed and fine-tuned. Explainable AI techniques such as LIME and LDA were used to interpret model decisions. The hybrid CNN-LSTM model with FastText embeddings achieved the highest performance with accuracy and F1-scores of 0.

Human Cognition-Based Consistency Inference Networks for Multi-Modal Fake News Detection (2023)

Existing multi-modal fake news detection models fail to capture inconsistent features between modalities, despite focusing on common semantics, which limits their ability to mimic human cognitive processes of detecting fake news through inconsistency analysis. We propose Human Cognition-based Consistency Inference Networks (HCCIN), which consists of three layers: a

cross-modal alignment layer to learn consistent semantics between text and image, a comment clue discovery layer to identify audience-concerned semantics from comments, and a collaborative inference layer to reason about consistency and inconsistency between news content and comments. Experiments on three public datasets demonstrate that HCCIN outperforms existing methods in multi-modal fake news detection, showing superior performance by effectively leveraging both consistent and inconsistent semantic cues.

A systematic review of multimodal fake news detection on social media using deep learning models (2025)

The rapid growth of unreliable information on social media necessitates effective methods for detecting multimodal fake news, which combines text, images, videos, audio, and user profiles. A systematic literature review (SLR) was conducted on studies published between 2018 and 2025, analyzing 963 articles and selecting 121 for in-depth review based on abstract and full-text eligibility criteria. The review focused on deep learning models used for multimodal fake news detection. Transformer models and Recurrent Neural Networks (RNNs) are the most widely used deep learning techniques for multimodal fake news detection, followed by Convolutional Neural Networks (CNNs).

[EMERGING] Large Language Models as Advisors

Bad Actor, Good Advisor: Exploring the Role of Large Language Models in Fake News Detection (2024)

The challenge of detecting fake news due to the need for diverse clues and real-world background knowledge, which current small language models (SLMs) struggle with, and the underexplored potential of large language models (LLMs) in this domain. An empirical study comparing GPT-3.5 with fine-tuned BERT, followed by the design of an adaptive rationale guidance network (ARG) that enables SLMs to selectively use multi-perspective rationales from LLMs, and a rationale-free distillation version (ARG-D) for cost-sensitive applications. ARG and ARG-D outperform three types of baseline methods: SLM-based, LLM-based, and hybrid models combining small and large language models.

[EMERGING] Graph-Based and Social Context Modeling

Bayesian Graph Local Extrema Convolution with Long-tail Strategy for Misinformation Detection (2024)

Detecting misinformation on social media, particularly addressing the challenge of spam bots retweeting false information and the over-concentration on high-degree nodes in graph neural networks. A Bayesian Graph Local Extrema Convolution (BLC) model is proposed for feature aggregation in graph structures, incorporating uncertainty and unreliable relationships. A long-tail strategy is introduced to better match long-tail users with the global social network, improving detection performance. The proposed long-tail strategy significantly improves the effectiveness of existing graph-based methods in detecting misinformation.

Constructing a User-Centered Fake News Detection Model by Using Classification Algorithms in Machine Learning Techniques (2023)

The rapid spread of fake news on social media necessitates effective automated detection systems, but existing methods often overlook user behavior, social network dynamics, and social capital in favor of linguistic and compositional features. The study developed a user-centered fake news detection model using machine learning techniques. It employed XGBoost to assess feature importance, identified key variables, and trained multiple classification models (SVM, RF, LR, CART, NN). Cross-validation was used to ensure model generalization and prevent overfitting or underfitting. The Random Forest (RF) model achieved the highest prediction accuracy at approximately 94%, while the Neural Network (NN) model had the lowest performance at about 92.

Research Gaps

[HIGH] Scarcity of large-scale, high-quality multimodal datasets, particularly for short video platforms and multilingual content, limits the development and benchmarking of robust detection models.

- Develop and open-source large-scale multimodal datasets for underrepresented languages and short-video formats.
- Create synthetic data generation techniques to augment existing datasets with realistic cross-modal inconsistencies.

[HIGH] Limited robustness of current models in real-world, dynamic scenarios where data is noisy, perturbed, or subject to adversarial attacks designed to evade detection.

- Design adversarial training frameworks specifically for multimodal fake news detectors.
- Investigate continual learning methods that allow models to adapt to evolving misinformation narratives without catastrophic forgetting.

[MEDIUM] Insufficient integration of temporal dynamics and propagation patterns in detection models, which often treat news as static artifacts rather than evolving stories within a social network.

- Model the temporal evolution of news stories and their associated user interactions using dynamic graph neural networks.

- Incorporate time-series analysis of propagation cascades to identify early signals of misinformation.

[MEDIUM] Lack of unified, cognitive-inspired frameworks that seamlessly integrate explainability with detection, moving beyond post-hoc explanations to inherently interpretable models.

- Develop neuro-symbolic AI approaches that combine neural perception with symbolic reasoning for transparent fake news detection.

- Create models that generate human-readable, evidence-based justifications for their classifications as a core function.

Future Perspectives

- Develop a unified, open-source benchmark suite that includes diverse multimodal datasets, standardized evaluation metrics, and adversarial robustness tests to accelerate comparable research.
- Design a real-time, multilingual fake news detection system for short-video platforms by integrating lightweight transformer models with cross-modal consistency checks and social context analysis.
- Create a human-in-the-loop framework where LLM-generated rationales are refined by fact-checkers to continuously train and improve smaller, deployable detection models.
- Investigate the fusion of graph-based propagation analysis with cognitive-inspired inconsistency detection to build a model that understands both 'what' is said and 'how' it spreads.
- Pioneer the use of causal inference methods in fake news detection to move beyond correlation-based features and identify the true causal drivers of misinformation spread and belief.

2. Global Synthesis (after 1 iteration(s))

Overview

The field of fake news detection has rapidly evolved to address the proliferation of misinformation across text, images, and video on social media. Deep learning dominates current approaches, with Transformer-based models (e.g., BERT, RoBERTa) and hybrid architectures (e.g., CNN-LSTM) achieving state-of-the-art results, often exceeding 98% accuracy or F1-score on benchmark datasets. Research is increasingly multimodal, integrating text, visual, and social context features to capture cross-modal inconsistencies that signal falsity. Explainable AI (XAI) and cognitive-inspired frameworks are emerging to improve model interpretability and mimic human reasoning. Key challenges remain in handling real-world data perturbations, multilingual content, and the scarcity of large-scale multimodal datasets, particularly for short videos.

Main Approaches

- Transformer-based models (e.g., BERT, RoBERTa, CT-BERT) fine-tuned for text classification
- Hybrid deep learning architectures combining CNNs, RNNs/LSTMs, and attention mechanisms
- Multimodal fusion frameworks integrating text, image, and social context features
- Graph neural networks leveraging social network structure and user interactions
- Large language model (LLM) guided or distilled approaches for rationale-based detection
- User-centered and feature-engineering methods using classical ML (e.g., Random Forest, XGBoost)

Common Datasets

Twitter datasets • Weibo datasets • FakeNewsNet • PHEME • ISOT dataset • Fakeddit dataset

Key Findings

1. Transformer models consistently outperform traditional RNN-based and CNN-based frameworks in capturing semantic and syntactic nuances for fake news detection.
2. Hybrid CNN-LSTM models with FastText embeddings can achieve near-perfect accuracy (0.99) and F1-scores (0.97-0.99) on multiple benchmark datasets.
3. Integrating multimodal features (text, image, comments, user profiles) significantly improves detection performance over unimodal baselines.
4. Document embedding quality is more critical than classification model complexity for achieving high accuracy.
5. Incorporating sentiment and emotion features from news content and user comments boosts detection AUC to 96.77%.
6. Graph-based methods with long-tail strategies and Bayesian convolution improve robustness under data perturbation, maintaining performance with 15% noise.
7. Explainable AI techniques (LIME, LDA) and cognitive-inspired reasoning frameworks enhance model interpretability and trustworthiness.

Synthesis covers **15** article(s).

3. Top Articles (15 selected)

#	Title	Venue	Year	Cited	Score
1	A systematic review of multimodal fake news detection on social media using de...	Results in Engineering	2025	47	0.66
2	Bad Actor, Good Advisor: Exploring the Role of Large Language Models in Fake N...	Proceedings of the AAAI Conference on Arti	2024	182	0.66
3	Advancing Fake News Detection: Hybrid Deep Learning With FastText and Explaina...	IEEE Access	2024	136	0.65
4	An overview of fake news detection: From a new perspective	Fundamental Research	2024	56	0.65
5	A comprehensive survey on machine learning approaches for fake news detection	Multimedia Tools and Applications	2023	113	0.64
6	It's All in the Embedding! Fake News Detection Using Document Embeddings	Mathematics	2023	82	0.64
7	Bootstrapping Multi-View Representations for Fake News Detection	Proceedings of the AAAI Conference on Arti	2023	105	0.64
8	Fake News Detection Model on Social Media by Leveraging Sentiment Analysis of ...	Sensors	2023	86	0.64
9	Multi-modal Misinformation Detection: Approaches, Challenges and Opportunities	ACM Computing Surveys	2024	52	0.64
10	FakeSV: A Multimodal Benchmark with Rich Social Context for Fake News Detectio...	Proceedings of the AAAI Conference on Arti	2023	94	0.64
11	Towards COVID-19 fake news detection using transformer-based models	Knowledge-Based Systems	2023	103	0.64
12	Bayesian Graph Local Extrema Convolution with Long-tail Strategy for Misinform...	ACM Transactions on Knowledge Discovery fr	2024	44	0.64
13	Human Cognition-Based Consistency Inference Networks for Multi-Modal Fake News...	IEEE Transactions on Knowledge and Data En	2023	78	0.64
14	SA-Bi-LSTM: Self Attention With Bi-Directional LSTM-Based Intelligent Model fo...	IEEE Access	2024	39	0.64
15	Constructing a User-Centered Fake News Detection Model by Using Classification...	IEEE Access	2023	59	0.63

Score Breakdown (venue / authors / impact / relevance)

#	Title (short)	Venue	Authors	Impact	Relev.	Final
1	A systematic review of multimodal fake new...	0.50	0.30	0.81	0.81	0.66
2	Bad Actor, Good Advisor: Exploring the Rol...	0.50	0.30	1.00	0.83	0.66
3	Advancing Fake News Detection: Hybrid Deep...	0.50	0.30	0.98	0.78	0.65
4	An overview of fake news detection: From a...	0.50	0.30	0.76	0.86	0.65
5	A comprehensive survey on machine learning...	0.50	0.30	0.86	0.84	0.64
6	It's All in the Embedding! Fake News Detec...	0.50	0.30	0.78	0.83	0.64
7	Bootstrapping Multi-View Representations f...	0.50	0.30	0.84	0.80	0.64

#	Title (short)	Venue	Authors	Impact	Relev.	Final
8	Fake News Detection Model on Social Media ...	0.50	0.30	0.79	0.82	0.64
9	Multi-modal Misinformation Detection: Appr...	0.50	0.30	0.74	0.70	0.64
10	FakeSV: A Multimodal Benchmark with Rich S...	0.50	0.30	0.81	0.78	0.64
11	Towards COVID-19 fake news detection using...	0.50	0.30	0.84	0.77	0.64
12	Bayesian Graph Local Extrema Convolution w...	0.50	0.30	0.70	0.70	0.64
13	Human Cognition-Based Consistency Inferenc...	0.50	0.30	0.77	0.76	0.64
14	SA-Bi-LSTM: Self Attention With Bi-Directi...	0.50	0.30	0.67	0.79	0.64
15	Constructing a User-Centered Fake News Det...	0.50	0.30	0.70	0.83	0.63

4. Article Summaries (15 articles) [Structured 6-field]

[1] A systematic review of multimodal fake news detection on social media using deep learnin...

Results in Engineering · 2025

Problem	The rapid growth of unreliable information on social media necessitates effective methods for detecting multimodal fake news, which combines text, images, videos, audio, and user profiles.
Method	A systematic literature review (SLR) was conducted on studies published between 2018 and 2025, analyzing 963 articles and selecting 121 for in-depth review based on abstract and full-text eligibility criteria. The review focused on deep learning models used for multimodal fake news detection.
Dataset	Twitter and Weibo datasets
Results	Transformer models and Recurrent Neural Networks (RNNs) are the most widely used deep learning techniques for multimodal fake news detection, followed by Convolutional Neural Networks (CNNs). The most common evaluation metrics are accuracy, precision, recall, and F-scores.
Contributions	Comprehensive analysis of deep learning models for multimodal fake news detection. • Identification of the most effective models (Transformers, RNNs, CNNs) and datasets (Twitter, Weibo). • Highlighting key evaluation metrics used in the field. • Outlining future research directions, including multilingual systems, hybrid models, explainable AI (XAI), and real-time detection.

[2] Bad Actor, Good Advisor: Exploring the Role of Large Language Models in Fake News Detect...

Proceedings of the AAAI Conference on Ar · 2024

Problem	The challenge of detecting fake news due to the need for diverse clues and real-world background knowledge, which current small language models (SLMs) struggle with, and the underexplored potential of large language models (LLMs) in this domain.
Method	An empirical study comparing GPT-3.5 with fine-tuned BERT, followed by the design of an adaptive rationale guidance network (ARG) that enables SLMs to selectively use multi-perspective rationales from LLMs, and a rationale-free distillation version (ARG-D) for cost-sensitive applications.
Dataset	Two real-world datasets
Results	ARG and ARG-D outperform three types of baseline methods: SLM-based, LLM-based, and hybrid models combining small and large language models.
Contributions	Demonstrating that LLMs, despite strong reasoning, underperform SLMs in fake news detection due to poor rationale integration. • Proposing LLMs as advisors to SLMs by providing multi-perspective rationales. • Introducing ARG, an adaptive rationale guidance network, and ARG-D, a distillation-based rationale-free variant, both showing superior performance over baselines.

[3] Advancing Fake News Detection: Hybrid Deep Learning With FastText and Explainable AI

IEEE Access · 2024

Problem	The widespread propagation of misinformation on social media platforms poses a significant challenge, requiring robust detection solutions that can effectively identify media-rich fake news by leveraging multimedia elements and enhancing detection accuracy.
Method	A hybrid deep learning approach integrating FastText word embeddings with Machine Learning and Deep Learning models, including Convolutional Neural Networks and Long Short-Term Memory networks, enhanced with regularization and hyperparameter optimization. Transformer-based models (BERT, XLNet, RoBERTa) were also employed and fine-tuned. Explainable AI techniques such as LIME and LDA were used to interpret model decisions.
Dataset	WELFake, FakeNewsNet, FakeNewsPrediction
Results	The hybrid CNN-LSTM model with FastText embeddings achieved the highest performance with accuracy and F1-scores of 0.99, 0.97, and 0.99 across all datasets. Transformer models outperformed traditional RNN-based frameworks in handling syntactic and semantic nuances. Explainable AI methods provided insights into the model's decision-making process.
Contributions	Introduction of a hybrid CNN-LSTM model with FastText embeddings for superior fake news detection performance. • Demonstration of transformer-based models' effectiveness in capturing semantic and syntactic features. • Integration of explainable AI techniques (LIME and LDA) to enhance model interpretability and trustworthiness. • Use of three diverse public datasets to validate model robustness across different contexts.

[4] An overview of fake news detection: From a new perspective		Fundamental Research · 2024
Problem	The rapid spread of fake news on the internet poses significant negative impacts on societies, countries, and individuals, necessitating effective detection methods to verify the veracity of news content.	
Method	A comprehensive survey analyzing fake news through three intrinsic characteristics—intentional creation, heteromorphic transmission, and controversial reception—used to categorize and evaluate existing detection approaches.	
Dataset	—	
Results	The review identifies key patterns in fake news propagation and provides a structured understanding of detection methods based on the three characteristics, offering insights into current techniques and future research directions.	
Contributions	Introduces a novel framework based on three intrinsic characteristics of fake news: intentional creation, heteromorphic transmission, and controversial reception. • Provides a systematic review of existing detection methods aligned with these characteristics. • Outlines future research trends and potential directions in fake news detection technology.	

[5] A comprehensive survey on machine learning approaches for fake news detection		Multimedia Tools and Applications · 2023
Problem	The proliferation of fake news on social media platforms poses significant societal and individual challenges, necessitating effective automatic fake news detection (FND) mechanisms to mitigate adverse consequences.	
Method	A comprehensive survey of machine learning (ML) and deep learning (DL) approaches for fake news detection, analyzing content-based, context-based, and hybrid-based features, along with definitions, psychological theories, datasets, and methodologies used in prior studies.	
Dataset	—	
Results	The survey provides a detailed analysis of ML/DL techniques, identifies key feature categories, summarizes commonly used datasets, and outlines challenges in current FND research, offering a structured overview to guide future work.	
Contributions	Comprehensive review of definitions and psychological underpinnings of fake news. • Systematic categorization of ML/DL methods based on content, context, and hybrid features. • Compilation of commonly used datasets and methodologies in FND research. • Identification of open challenges and future research directions in fake news detection.	

[6] It's All in the Embedding! Fake News Detection Using Document Embeddings		Mathematics · 2023
Problem	The increasing spread of fake news on social media platforms, which undermines public trust, polarizes society, and manipulates opinions on critical issues such as elections and vaccinations, necessitates effective automated detection methods.	
Method	The study proposes a novel approach using document embeddings to construct multiple classification models for fake news detection. It evaluates different architectures for binary and multi-labeled classification, focusing on the impact of document encoding on model performance.	
Dataset	Five large news corpora	
Results	The proposed models achieved higher accuracy than more complex state-of-the-art deep neural network models. The study found that the quality of document encoding is the most critical factor for high accuracy, surpassing the complexity of the classification model.	
Contributions	Demonstrated that document embedding quality is more important than model complexity for fake news detection. • Provided a benchmark of multiple architectures using document embeddings for binary and multi-labeled classification. • Achieved superior performance compared to complex deep learning models using simpler, embedding-focused approaches.	

[7] Bootstrapping Multi-View Representations for Fake News Detection		Proceedings of the AAAI Conference on Ar · 2023
Problem	Understanding how cross-modal consistency relates to news fidelity and how features from different modalities affect decision-making in fake news detection.	
Method	Bootstrapping Multi-view Representations (BMR) using improved Multi-gate Mixture-of-Expert networks (iMMoE) for feature refinement and fusion. Each modality (text, image pattern, image semantics) is used to predict news fidelity and cross-modal consistency, with reweighting based on prediction scores for bootstrapping.	
Dataset	—	
Results	BMR outperforms state-of-the-art methods on typical fake news detection datasets, demonstrating improved accuracy in fake news detection through effective multi-view representation bootstrapping.	
Contributions	Introduction of a novel BMR framework for fake news detection. • Use of iMMoE for refined feature fusion across modalities. • Integration of cross-modal consistency prediction to guide representation bootstrapping. • Demonstrated superior performance over existing methods on benchmark datasets.	

[8] Fake News Detection Model on Social Media by Leveraging Sentiment Analysis of News Conte...		Sensors · 2023
Problem	The spread of fake news on social media platforms poses significant threats to political, economic, and social stability, necessitating effective detection methods.	
Method	A bidirectional long short-term memory (Bi-LSTM) model was proposed, incorporating features derived from sentiment analysis of news content and emotion analysis of user comments to detect fake news.	
Dataset	Fakeddit dataset	
Results	The proposed model achieved a detection accuracy of 96.77% as measured by the Area Under the ROC Curve (AUC), outperforming existing state-of-the-art methods.	
Contributions	Integration of sentiment analysis of news content and emotion analysis of user comments as discriminative features. • Development of a Bi-LSTM model that effectively leverages both content and user reaction features. • Demonstration of improved detection performance with an AUC of 96.77%, surpassing previous approaches.	

[9] Multi-modal Misinformation Detection: Approaches, Challenges and Opportunities		ACM Computing Surveys · 2024
Problem	The increasing prevalence of multi-modal misinformation on social media platforms, where false information is spread through deceptive connections between text and visual content such as images and videos.	
Method	A comprehensive analysis and categorization of existing automatic techniques for detecting cross-modal discordance in multi-modal web content, focusing on the integration of textual and visual modalities.	
Dataset	—	
Results	The study identifies key approaches in multi-modal misinformation detection, highlights challenges such as semantic misalignment and context dependency, and reveals research opportunities in improving cross-modal understanding and detection accuracy.	
Limitations	Existing methods often lack robustness in real-world scenarios due to limited datasets, insufficient modeling of contextual nuances, and challenges in handling diverse multimodal content.	
Contributions	Systematic categorization of multi-modal misinformation detection approaches. • Identification of critical challenges including cross-modal alignment and contextual understanding. • Proposal of new research directions to enhance detection systems through improved multimodal integration and contextual awareness.	

[10] FakeSV: A Multimodal Benchmark with Rich Social Context for Fake News Detection on Short...		Proceedings of the AAAI Conference on Ar · 2023
Problem	The scarcity of comprehensive and large-scale datasets and insufficient utilization of multimodal information hinder effective fake news detection on short video platforms.	
Method	Construction of the FakeSV dataset and development of the SV-FEND multimodal detection model that leverages cross-modal correlations and social context information.	
Dataset	FakeSV, the largest Chinese short video dataset for fake news detection, including news content, user comments, and publisher profiles.	
Results	Extensive experiments demonstrate the superiority of the SV-FEND model, providing detailed comparisons of different methods and modalities, and highlighting the importance of multimodal and social context features.	
Contributions	Creation of the largest Chinese short video dataset for fake news detection (FakeSV) with rich multimodal and social context data. • Introduction of the SV-FEND model that effectively exploits cross-modal correlations and social context for improved detection performance. • Comprehensive exploratory analysis of fake news characteristics in short videos. • Open-sourcing of the dataset and code to support future research.	
[11] Towards COVID-19 fake news detection using transformer-based models		Knowledge-Based Systems · 2023
Problem	The surge of fake news during the COVID-19 pandemic poses significant public health risks, particularly due to the mixing of true and false information, making detection challenging.	
Method	The study evaluates machine learning algorithms and fine-tunes pre-trained transformer models (BERT and CT-BERT) with downstream neural network structures (CNN and BiGRU) applied on top, using both frozen and unfrozen parameters.	
Dataset	A real-world COVID-19 fake news dataset	
Results	Incorporating BiGRU on top of the CT-BERT model achieved a state-of-the-art F1 score of 98%, demonstrating superior performance in detecting COVID-19 fake news.	
Contributions	Demonstrated the effectiveness of CT-BERT for detecting mixed true/false COVID-19 news. • Achieved a state-of-the-art F1 score of 98% using BiGRU on top of CT-BERT. • Provided insights into the impact of frozen vs. unfrozen parameters in transformer fine-tuning.	
[12] Bayesian Graph Local Extrema Convolution with Long-tail Strategy for Misinformation Dete...		ACM Transactions on Knowledge Discovery · 2024
Problem	Detecting misinformation on social media, particularly addressing the challenge of spam bots retweeting false information and the over-concentration on high-degree nodes in graph neural networks.	
Method	A Bayesian Graph Local Extrema Convolution (BLC) model is proposed for feature aggregation in graph structures, incorporating uncertainty and unreliable relationships. A long-tail strategy is introduced to better match long-tail users with the global social network, improving detection performance.	
Dataset	Two public Twitter datasets	
Results	The proposed long-tail strategy significantly improves the effectiveness of existing graph-based methods in detecting misinformation. BLC demonstrates robust performance, consistently outperforming traditional algorithms even when 15% of the data is perturbed.	
Contributions	Introduction of Bayesian Graph Local Extrema Convolution (BLC) to handle uncertainty and unreliable relationships in graph structures. • Proposal of a long-tail strategy to mitigate over-concentration on high-degree nodes in graph neural networks. • Demonstration of improved misinformation detection performance on real-world Twitter datasets. • Validation of BLC's robustness under data perturbation across three graph datasets.	

[13] Human Cognition-Based Consistency Inference Networks for Multi-Modal Fake News Detection		IEEE Transactions on Knowledge and Data · 2023
Problem	Existing multi-modal fake news detection models fail to capture inconsistent features between modalities, despite focusing on common semantics, which limits their ability to mimic human cognitive processes of detecting fake news through inconsistency analysis.	
Method	We propose Human Cognition-based Consistency Inference Networks (HCCIN), which consists of three layers: a cross-modal alignment layer to learn consistent semantics between text and image, a comment clue discovery layer to identify audience-concerned semantics from comments, and a collaborative inference layer to reason about consistency and inconsistency between news content and comments.	
Dataset	Weibo, Twitter, PHEME	
Results	Experiments on three public datasets demonstrate that HCCIN outperforms existing methods in multi-modal fake news detection, showing superior performance by effectively leveraging both consistent and inconsistent semantic cues.	
Contributions	Introducing a human cognition-inspired framework for fake news detection that explicitly models inconsistency between modalities. • Proposing a three-layer architecture that captures consistent semantics, audience-focused comment clues, and collaborative reasoning of consistency/inconsistency. • Demonstrating state-of-the-art performance on multiple benchmark datasets, validating the effectiveness of incorporating cognitive reasoning in multi-modal detection.	

[14] SA-Bi-LSTM: Self Attention With Bi-Directional LSTM-Based Intelligent Model for Accurate...		IEEE Access · 2024
Problem	The increasing spread of misinformation on social media platforms necessitates accurate and reliable fake news detection to ensure information integrity and public trust.	
Method	A novel Self-Attention with Bi-Directional LSTM (SA-BiLSTM) model is proposed, combining the sequential pattern recognition capabilities of Bi-LSTM with a self-attention mechanism to focus on critical text segments for improved discrimination between real and fake news.	
Dataset	ISOT dataset, consisting of diverse news articles annotated by human fact-checkers as real or fake.	
Results	The SA-BiLSTM model achieved 99.98% predictive accuracy on test data, outperforming baseline models such as CNN, GRU, Attention-GRU, LSTM, and Bi-LSTM, and demonstrating competitive results against state-of-the-art methods.	
Contributions	Introduction of a self-attention mechanism integrated with Bi-LSTM to enhance focus on salient text features. • Demonstration of superior performance (99.98% accuracy) on the ISOT dataset. • Ablation studies confirming the significant impact of the attention mechanism on model performance. • A robust framework for real-time fake news detection to support information integrity on social media.	

[15] Constructing a User-Centered Fake News Detection Model by Using Classification Algorithm...		IEEE Access · 2023
Problem	The rapid spread of fake news on social media necessitates effective automated detection systems, but existing methods often overlook user behavior, social network dynamics, and social capital in favor of linguistic and compositional features.	
Method	The study developed a user-centered fake news detection model using machine learning techniques. It employed XGBoost to assess feature importance, identified key variables, and trained multiple classification models (SVM, RF, LR, CART, NNET). Cross-validation was used to ensure model generalization and prevent overfitting or underfitting.	
Dataset	—	
Results	The Random Forest (RF) model achieved the highest prediction accuracy at approximately 94%, while the Neural Network (NNET) model had the lowest performance at about 92.1%. The results indicate that incorporating user, content, and social network characteristics improves detection accuracy.	
Contributions	Proposed a user-centered fake news detection model integrating user behavior, content, and social network features. • Utilized XGBoost to identify the most influential variables affecting fake news detection. • Demonstrated superior performance of the Random Forest model in detecting fake news compared to other machine learning classifiers. • Enhanced model robustness through cross-validation to prevent overfitting and improve generalization.	

5. Query Expansion (4 queries)

The **QueryExpander** agent produces diverse search queries from the original topic, maximising coverage of different angles, synonyms, and sub-tasks. OpenAlex is searched independently for each query; results are then deduplicated.

1. fake news detection [original]
2. misinformation detection [variant 1]
3. machine learning fake news [variant 2]
4. fake news detection social media [variant 3]

6. Reflexion Loop (1 iteration(s), 1 critique(s))

Config: **MAX_REFLEXION_ITERATIONS = 3** — **REFLEXION_MIN_QUALITY = GOOD**. The Critic evaluates the Synthesis on 4 axes: *fidelity*, *completeness*, *specificity*, *consistency*. If quality is below the threshold, it requests a revision.

Iteration 1 — Quality: **GOOD** — **APPROVED**

Issues identified by Critic:

- The synthesis claims 'Transformer-based models (e.g., BERT, RoBERTa) and hybrid architectures (e.g., CNN-LSTM) achieving state-of-the-art results, often exceeding 98% accuracy or F1-score on benchmark datasets.' This conflates two separate findings: the 98% F1 is from CT-BERT+BiGRU on a COVID-19 dataset (openalex_W4377115522), while the 0.99 accuracy/F1 is from a hybrid CNN-LSTM with FastText on WELFake/FakeNewsNet (openalex_W4393170771). The synthesis implies these metrics are general across benchmarks, which is misleading.
- The synthesis states 'Document embedding quality is more critical than classification model complexity for achieving high accuracy' as a key finding, but does not mention that this conclusion was drawn from a study using simpler models on five large news corpora (openalex_W4317505581), which may not generalize to all contexts.
- The synthesis lists 'Fakeddit dataset' under common_datasets, but only one summary (openalex_W4319320113) uses it, and it is not mentioned as a common benchmark in other summaries. This overstates its prevalence.
- The synthesis mentions 'Graph neural networks leveraging social network structure and user interactions' as a main approach, but the only graph-based method in the summaries is Bayesian Graph Local Extrema Convolution (BLC) with a long-tail strategy (openalex_W4390547902). The synthesis does not capture the specific Bayesian and long-tail innovations.
- The synthesis claims 'Explainable AI techniques (LIME, LDA) and cognitive-inspired reasoning frameworks enhance model interpretability and trustworthiness' as a key finding, but the cognitive-inspired framework (HCCIN from openalex_W4378697009) is about detection performance, not interpretability. This conflates XAI with cognitive-inspired detection.

Suggestions given to Synthesizer:

- Clarify that the 98% F1 and 0.99 accuracy/F1 are from specific models on specific datasets (CT-BERT+BiGRU on COVID-19 data; CNN-LSTM+FastText on WELFake/FakeNewsNet) rather than implying they are general benchmarks.
- Add context that the finding on document embedding quality vs. model complexity is based on a study using simpler models and five large corpora, and may not apply to all detection scenarios.
- Remove 'Fakeddit dataset' from common_datasets or replace it with datasets mentioned in multiple summaries (e.g., Twitter, Weibo, FakeNewsNet).
- Specify the graph-based approach as 'Bayesian Graph Local Extrema Convolution with long-tail strategies' to accurately reflect the method from the summaries.
- Separate the XAI finding from the cognitive-inspired reasoning finding: state that XAI techniques (LIME, LDA) improve interpretability, while cognitive-inspired frameworks (HCCIN) improve detection by modeling cross-modal inconsistency.

Final: Synthesizer completed 1 iteration(s). Maximum allowed: 3. Quality threshold required: **GOOD**.

7. Agent Execution Logs (7 agents)

Agent	Status	Duration (s)	API Calls	Tokens
QueryExpander	OK	28.48	4	3,730
Searcher	OK	6.22	4	0
QualityCritic	OK	43.69	0	0
Summarizer	OK	74.62	15	15,046
Synthesizer	OK	8.90	1	5,286
Critic	OK	10.99	1	6,014
TrendAnalyst	OK	18.46	1	6,967
TOTAL	—	—	—	37,043